

Statistical modeling

Module 01

Marc-Olivier Boldi

Bachelor, GSEM

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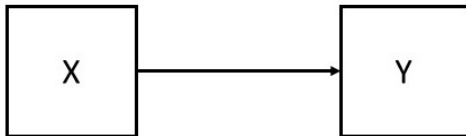
Agenda

- ▶ What is a statistical model?
- ▶ Reminders of univariate statistics
- ▶ Correlation
- ▶ Other dependence concepts

Introduction: What is a model?

A model is a **simplified and imperfect** representation of the reality.

In this course, a link description between a response Y and some characteristics X .



Introduction: What is a model?

Examples of y explained by x 's:

- ▶ Supply chain: demand explained by time (season), special events, other variables.
- ▶ Banking: loan refunding explained by personal customer characteristics.
- ▶ Marketing: product buying chance explained by previous buying habits, personal characteristics...
- ▶ HR: burn out risk explained by personal characteristics.
- ▶ Psychology: behavior explained by personal preferences.
- ▶ etc., etc., etc. (almost every management and scientific fields)

Introduction: modeling challenges

Model specification: to be able to define the "real" relationship between variables x and y .

For example, the attendance to a music festival may increase with temperature forecast for moderate temperature range and then decrease for very high temperatures. A useful model should take that up-down effect of the weather.

Variables selection: to be able to discover and select useful x 's.

For example, the attendance may be explained by the price of the butter in the model but this is useless. On the other hand, it could be explained by the attendance to another festival happening just before. However, that information is not available.

Introduction: What is a **statistical** model?

A model built from the observation of a sample $\{y_i, x_i\}_{i=1}^n$ of individuals, cases, subjects, occurrences, etc.

Statistical challenge: partial observation of the population.

For example, the model aims at explaining the expenses of a customer in a supermarket according to its characteristics.

However, only 100 people (the sample) are in the study (in opposition to the whole population).

For example, the model aims at predicting the demand in the future for a given product. However, only past demands can be observed (of course...).

Statistical results aim at being extrapolated to the whole (unobserved) population; a difficult task.

Some reminders of univariate statistics

In statistic, observed data constitute the sample. Unobserved data constitute the population (of interest).

Most used data type:

- ▶ Numerical:
 - ▶ Continuous: possible results can't be counted (e.g. time to event, height, spending, etc.)
 - ▶ Discrete: possible results can be counted (e.g. number of people waiting in line, calls to a call center, etc.)
- ▶ Categorical: possible results are categories
 - ▶ Nominal: no order (e.g. sex, spoken language, etc.)
 - ▶ Ordinal: ordered (e.g. level of education, level of satisfaction, etc.)

Some reminders of univariate statistics

Statistical notions for numerical observation

Sample	Population
Observation y_i	Random variable Y
Sample size n	Pop. size N , ∞ , or continuous
Mean $\bar{y} = \frac{1}{n} \sum_i y_i$	Expectation $E(Y)$
Variance $s^2 = \frac{1}{n-1} \sum_i (y_i - \bar{y})^2$	$Var(Y) = E((Y - E(Y))^2)$
Standard deviation $s = \sqrt{s^2}$	$SD(Y) = \sqrt{Var(Y)}$

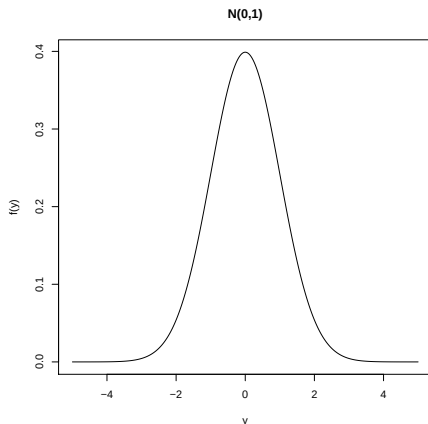
Object describing the distribution

Type	Sample	Population
Num. continuous	Histogram	Density function
Num. discrete	Probability table	Probability function
Categorical	Probability table	Probability function

Some reminders of univariate statistics

Standard normal distribution $Y \sim N(0, 1)$

$$f(y) = \frac{1}{\sqrt{2\pi}} \exp\left(-y^2/2\right), \quad E(Y) = 0, \quad \text{Var}(Y) = 1$$



The 68 – 95 – 99.7 rule:

- ▶ $P(-1 \leq Y \leq 1) = 0.68$
- ▶ $P(-2 \leq Y \leq 2) = 0.95$
- ▶ $P(-3 \leq Y \leq 3) = 0.997$

Some reminders of univariate statistics

Normal distribution $Y \sim N(\mu, \sigma^2)$

$$f(y) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{(y - \mu)^2}{2\sigma^2} \right\}, \quad E(Y) = \mu, \quad \text{Var}(Y) = \sigma^2$$

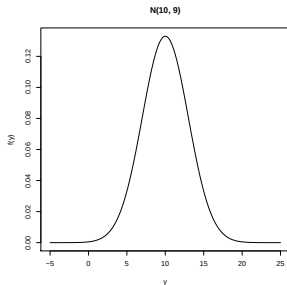
Link between $N(0, 1)$ and $N(\mu, \sigma^2)$:

$$Y = \mu + \sigma Z, \quad Z = \frac{Y - \mu}{\sigma}$$

where $Z \sim N(0, 1)$.

The 68 – 95 – 99.7 rule:

- ▶ $P(-\sigma \leq Y \leq \sigma) = 0.68$
- ▶ $P(-2\sigma \leq Y \leq 2\sigma) = 0.95$
- ▶ $P(-3\sigma \leq Y \leq 3\sigma) = 0.997$



Some reminders of univariate statistics

For $Z \sim N(0, 1)$:

$$P(Z \leq q) = p$$

Computation using R,

- ▶ Density: for given x , $f(x)$ is `dnorm(x, mean=0, sd=1)`.
- ▶ Probability: for given q , p =`pnorm(q, mean=0, sd=1)`.
- ▶ Quantile: for given p , q =`qnorm(p, mean=0, sd=1)`.
- ▶ Common notation of the quantile: z_p .

Some reminders of univariate statistics

Student t distribution with df degrees of freedom $Y \sim t_{df}$.

$$E(Y) = 0, \quad \text{Var}(Y) = \frac{df}{df - 2}.$$

Density function is complex.

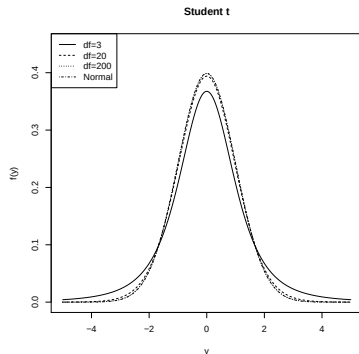
In R, Density: `dt(x, df)`.

Probability: for $t_{df}(q)$, use `pt(q, df)`

Quantile: for $t_{df;p}$, use `qt(p, df)`

Approximation by the normal:

When $df \rightarrow \infty$, t_{df} tends to $N(0, 1)$.



Some reminders of univariate statistics

Chi-square χ^2 distribution with df degrees of freedom $Y \sim \chi^2_{df}$.

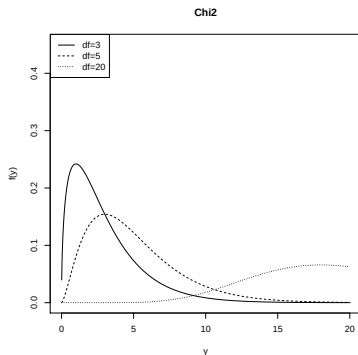
$$E(Y) = df, \quad \text{Var}(Y) = df^2$$

Density function is complex.

In R, Density: `dchisq(x, df)`.

Probability: for $\chi^2_{df}(q)$, use `pchisq(q, df)`

Quantile: for $\chi^2_{df;p}$, use `qchisq(p, df)`



Some reminders of univariate statistics

Fisher F distribution with df_1 and df_2 degrees of freedom

$$Y \sim F_{df_1, df_2}.$$

$$E(Y) = \frac{df_2}{df_2 - 2},$$

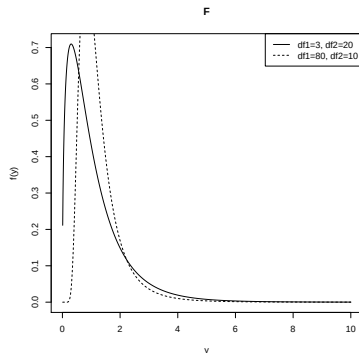
$$Var(Y) = \frac{2df_2^2(df_1 + df_2 - 2)}{df_1(df_2 - 2)^2(df_2 - 4)}$$

Density function is complex.

In R, Density: `df(x, df1, df2)`.

Probability: for $F_{df_1, df_2}(q)$, use `pf(q, df1, df2)`

Quantile: for $F_{df_1, df_2; p}$, use `qf(p, df1, df2)`



Some reminders of univariate statistics

Some links between these distributions:

- ▶ If $Y \sim N(0, 1)$ then $Y^2 \sim \chi_1^2$.
- ▶ If $Y_1, \dots, Y_n \sim N(0, 1)$, independent, then $\sum_{i=1}^n Y_i^2 \sim \chi_n^2$
- ▶ If $Y \sim N(0, 1)$ and $V \sim \chi_v^2$, independent, then

$$\frac{Y}{\sqrt{V/v}} \sim t_v$$

- ▶ If $V \sim \chi_v^2$ and $W \sim \chi_w^2$, independent, then

$$\frac{V/v}{W/w} \sim F_{v,w}.$$

Some reminders of univariate statistics

Point estimate of the mean (sample estimate): $\bar{y}$

Confidence interval at level $1 - \alpha$ (population estimate):

$$\left[\bar{y} - t_{n-1; 1-\alpha/2} s / \sqrt{n}, \bar{y} + t_{n-1; 1-\alpha/2} s / \sqrt{n} \right]$$

Approximate (normal) confidence interval:

$$\left[\bar{y} - z_{1-\alpha/2} s / \sqrt{n}, \bar{y} + z_{1-\alpha/2} s / \sqrt{n} \right]$$

Valid if n is large.

α	$z_{1-\alpha/2}$
0.40	0.84
0.30	1.04
0.20	1.28
0.10	1.64
0.05	1.96

In R, $z_{1-\alpha/2}$ is `qnorm(p=1-a/2)`.

Some reminders of univariate statistics

Bilateral t-test (two-sided):

$$H_0 : E(Y) = m_0 \quad \text{versus} \quad H_1 : E(Y) \neq m_0$$

Statistic:

$$t = \frac{\bar{y} - m_0}{s/\sqrt{n}}$$

Distribution if H_0 is true: t_{n-1} (Student with $\text{df}=n-1$).

p -value:

$$p = 2 - 2 \times t_{n-1}(t).$$

Test at level α : reject H_0 if (equiv.)

- ▶ $|t| > t_{n-1;1-\alpha/2}$,
- ▶ $p < \alpha$.

Some reminders of univariate statistics

Bilateral t-test (one-sided):

$$H_0 : E(Y) = m_0 \quad \text{versus} \quad H_1 : E(Y) > m_0 \quad [<]$$

Statistic:

$$t = \frac{\bar{y} - m_0}{s/\sqrt{n}}$$

Distribution if H_0 is true: t_{n-1}

p -value:

$$p = 1 - t_{n-1}(t) \quad [p = t_{n-1}(t)]$$

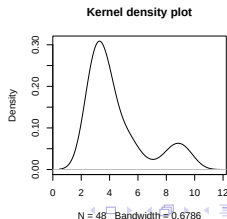
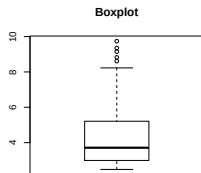
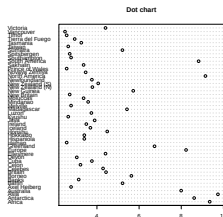
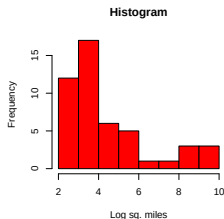
Test at level α : reject H_0 if (equiv.)

- ▶ $t > t_{n-1;1-\alpha}, \quad [t < t_{n-1;\alpha}]$
- ▶ $p < \alpha.$

Some reminders of univariate statistics

Graphical representation of numerical data: histogram, dot chart, boxplot, kernel density plot

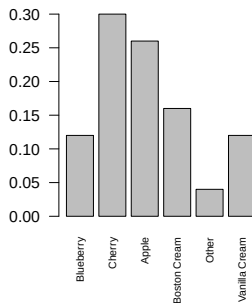
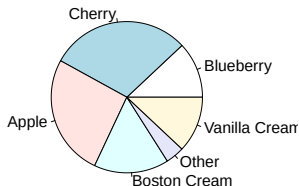
Example: world's areas ($\times 1000 \text{ mi}^2$; log) of the landmasses exceeding $10,000 \text{ mi}^2$.



Some reminders of univariate statistics

Graphical representation of categorical data: pie chart and bar plot

Example: Pie chart of pie sales.



Description of the dependence between two variables

Context: same subject i , two measures x_i, y_i .

How to describe their dependence?

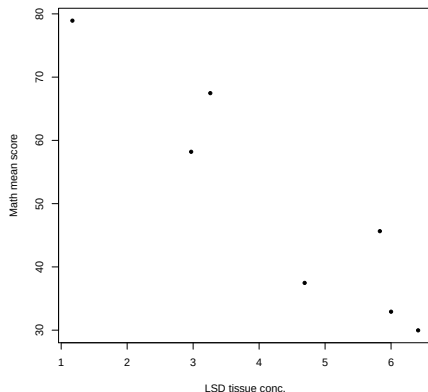
For examples:

- ▶ Height and weight of a subject: high/low together?
- ▶ Speed and consumption along a journey: high/low together?
- ▶ Happiness and productivity, gender and salary, age and

Dependence of two numerical variables: scatter plot

Graph of x versus y .

Example: LSD concentration in tissue versus a Math exam score



Dependence is negative and (approximately) linear.

Source: Wagner, Agahajanian, and Bing (1968). Correlation of Performance Test Scores with Tissue Concentration of Lysergic Acid Diethylamide in Human Subjects. Clinical Pharmacology and Therapeutics, Vol.9 pp635-638.

Dependence between two numerical variables: correlation

Correlation measures the linear dependence between x and y unconditionally on the other variables

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

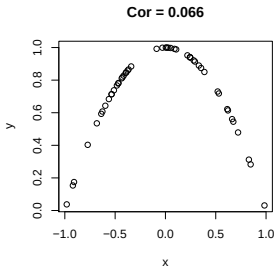
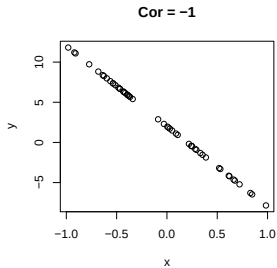
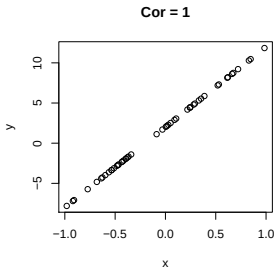
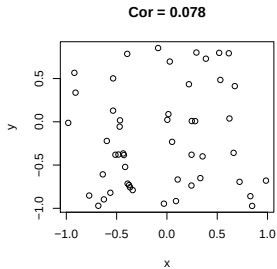
Correlation is between -1 (strong negative) and 1 (strong positive).

In the example: $r_{xy} = -0.94$

```
> cor(lsd)
```

	LSD	Math
LSD	1.0000000	-0.9369285
Math	-0.9369285	1.0000000

Correlation measures linear dependence



Population correlation and correlation test

The correlation of the population (unobserved):

$$\text{Cor}(X, Y) = \frac{E[(X - E(X))(Y - E(Y))]}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

Test on the correlation

$$H_0 : \text{Cor}(X, Y) = 0 \quad \text{versus} \quad H_1 : \text{Cor}(X, Y) \neq 0$$

$$\text{Statistics under } H_0: t = \frac{r_{xy}}{\sqrt{(1-r_{xy}^2)/(n-2)}} \sim t_{n-2}$$

$$\text{Reject } H_0: |t| > t_{n-2; 1-\alpha/2}$$

$$\text{P-value: } p = 2 - 2 \times t_{n-2}(t)$$

Correlation test: example

```
> cor.test(x=lsd$LSD, y=lsd$Math)
```

Pearson's product-moment correlation

data: lsd\$LSD and lsd\$Math

t = -5.994, df = 5, p-value = 0.001854

alternative hypothesis: true correlation is not equal to 0

95 percent confidence interval:

-0.9908681 -0.6244782

sample estimates:

cor

-0.9369285

H_0 is rejected at 5% since the p -value is lower than 5%.

Covariance

Covariance is a non standardized version of the correlation:

- ▶ Sample covariance:

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

- ▶ Population covariance:

$$\text{Cov}(X, Y) = E\{(X - E(X))(Y - E(Y))\}.$$

- ▶ Variance: the covariance of a variable with itself is the variance

$$s_{xx} = s_x^2, \quad \text{Cov}(X, X) = \text{Var}(X).$$

Example:

```
> cov(lsd)
      LSD      Math
LSD    3.745824 -33.74787
Math -33.747874 346.36389
> cor(lsd)[1,2]*sd(lsd$Math)*sd(lsd$LSD)
[1] -33.74787
```

Units

Suppose that X is in meters m , and $x_1, \dots, x_n$ is a sample. Then

- ▶ $E(X)$ and $\bar{x}$ are in m .
- ▶ $SD(X)$ and s_x are in m .
- ▶ $Var(X)$ and s_y^2 are in m^2 .

Furthermore, suppose that another measure is taken, say Y is in g , and $y_1, \dots, y_n$ the sample, then

- ▶ $Cov(X, Y)$ and s_{xy} are in $m \times g$.
- ▶ $Cor(X, Y)$ and r_{xy} have no units.

Notations

Frequent notations use Greek letters:

- ▶ $E(X)$: μ_X or μ (if mention to X is not needed),
- ▶ $Var(X)$: σ_X^2 or σ^2 ,
- ▶ $SD(X)$: σ_X or σ ,
- ▶ $Cov(X, Y)$: σ_{XY} ,
- ▶ $Cor(X, Y)$: ρ_{XY} or ρ .

Covariance and correlation matrices

Extending to more than two variables, the covariance matrix is the matrix of covariances:

$$\Sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1p} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{p1} & \sigma_{p2} & \cdots & \sigma_{pp} \end{bmatrix}, \quad S = \begin{bmatrix} s_{11} & s_{12} & \cdots & s_{1p} \\ s_{21} & s_{22} & \cdots & s_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ s_{p1} & s_{p2} & \cdots & s_{pp} \end{bmatrix}$$

where

$$s_{jj'} = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{ij'} - \bar{x}_{j'}), \quad \sigma_{jj'} = E\{(X_j - E(X_j))(X_{j'} - E(X_{j'}))\}.$$

The diagonal elements of Σ and S are the variances (population and sample) of the variables

$$\sigma_{jj} = \text{Var}(X_j), \quad s_{jj} = s_j^2.$$

Covariance and correlation matrices

Example: Macroeconomic data set `longley` (see `?longley` for details).

```
> signif(cov(longley), digit=4)
```

	GNP.deflator	GNP	Unemployed	Armed.Forces	Population	Year	Employed
GNP.deflator	116.50	1064.0	625.9	349.0	73.50	50.92	36.80
GNP	1064.00	9879.0	5612.0	3088.0	685.20	471.00	343.30
Unemployed	625.90	5612.0	8732.0	-1154.0	446.30	297.30	164.90
Armed.Forces	349.00	3088.0	-1154.0	4843.0	176.40	138.20	111.80
Population	73.50	685.2	446.3	176.4	48.39	32.92	23.46
Year	50.92	471.0	297.3	138.2	32.92	22.67	16.24
Employed	36.80	343.3	164.9	111.8	23.46	16.24	12.33

```
> with(longley, var(GNP.deflator))  
[1] 116.4576
```

Covariance and correlation matrices

Correlation matrices are similar:

$$P = \begin{bmatrix} \rho_{11} & \rho_{12} & \cdots & \rho_{1p} \\ \rho_{21} & \rho_{22} & \cdots & \rho_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{p1} & \rho_{p2} & \cdots & \rho_{pp} \end{bmatrix}, \quad R = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1p} \\ r_{21} & r_{22} & \cdots & r_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ r_{p1} & r_{p2} & \cdots & r_{pp} \end{bmatrix}$$

where

$$r_{jj'} = \frac{s_{jj'}}{s_j s_{j'}}, \quad \rho_{jj'} = \frac{\sigma_{jj'}}{\sigma_j \sigma_{j'}}.$$

The diagonal elements of P and R are equal to 1.

Covariance and correlation matrices

Example:

```
> signif(cor(longley), digit=4)
```

	GNP.deflator	GNP	Unemployed	Armed.Forces	Population	Year	Employed
GNP.deflator	1.0000	0.9916	0.6206	0.4647	0.9792	0.9911	0.9709
GNP	0.9916	1.0000	0.6043	0.4464	0.9911	0.9953	0.9836
Unemployed	0.6206	0.6043	1.0000	-0.1774	0.6866	0.6683	0.5025
Armed.Forces	0.4647	0.4464	-0.1774	1.0000	0.3644	0.4172	0.4573
Population	0.9792	0.9911	0.6866	0.3644	1.0000	0.9940	0.9604
Year	0.9911	0.9953	0.6683	0.4172	0.9940	1.0000	0.9713
Employed	0.9709	0.9836	0.5025	0.4573	0.9604	0.9713	1.0000

Correlations, tests, and scatter plots

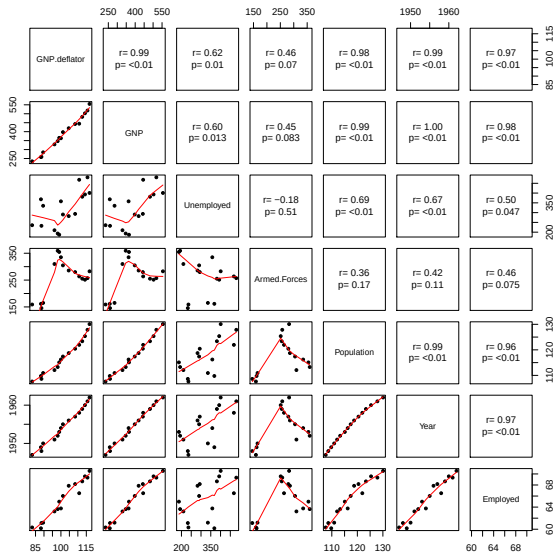
Combining these two tools¹ in R.

```
panel.cor <- function(x, y, digits = 2, cex.cor, ...)  
{  
  usr <- par("usr"); on.exit(par(usr))  
  par(usr = c(0, 1, 0, 1))  
  # correlation coefficient  
  r <- cor(x, y)  
  txt <- format(c(r, 0.123456789), digits = digits)[1]  
  txt <- paste("r= ", txt, sep = "")  
  text(0.5, 0.6, txt)  
  
  # p-value calculation  
  p <- cor.test(x, y)$p.value  
  txt2 <- format(c(p, 0.123456789), digits = digits)[1]  
  txt2 <- paste("p= ", txt2, sep = "")  
  if(p<0.01) txt2 <- paste("p= ", "<0.01", sep = "")  
  text(0.5, 0.4, txt2)  
}  
  
> pairs(longley, upper.panel = panel.cor, pch=20, lower.panel=panel.smooth)
```

\$

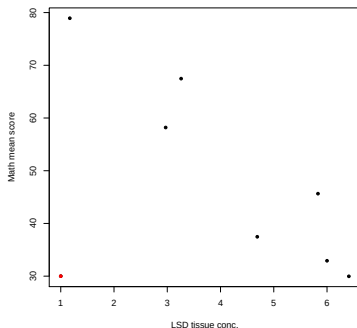
¹Source: <https://www.r-bloggers.com/scatter-plot-matrices-in-r/>

Correlations, tests, and scatter plots



(Un-)Robustness

The correlation is influenced by extreme values. A low math score, though no drug. The correlation goes down to -0.52 .



Several correlations exist: Pearson's correlation, Spearman's rho, Kendall's tau.

Spearman's ρ

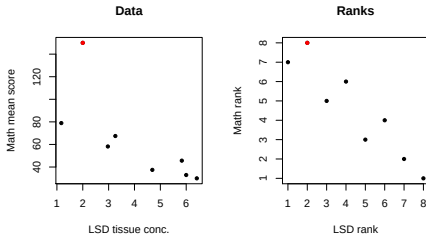
Instead of using the correlation of the observed values $(y_i, x_i)_{i=1}^n$, the correlation of the ranks is computed: an attempt to be more robust against extremes.

Example: LSD data.

LSD	LSD rank	Math rank	Math
1.17	1	7	78.93
2.00	2	8	150.00
2.97	3	5	58.20
3.26	4	6	67.47
4.69	5	3	37.47
5.83	6	4	45.65
6.00	7	2	32.92
6.41	8	1	29.97

Spearman's ρ

	Without extreme value	With extreme value
Pearson	-0.94	-0.75
Spearman	-0.93	-0.93



Spearman's ρ test

Similar to the correlation test

$$H_0 : \rho_S(X, Y) = 0 \quad \text{versus} \quad H_1 : \rho_S(X, Y) \neq 0$$

Statistics under H_0 : $t = r_S \sqrt{\frac{n-2}{1-r_S^2}} \sim t_{n-2}$.

Reject H_0 : $|t| > t_{n-2; 1-\alpha/2}$

P-value: $p = 2 - 2 \times t_{n-2}(t)$

Spearman's ρ test

```
> cor.test(x=lsd2$LSD, y=lsd2$Math, method="spearman")
```

Spearman's rank correlation rho

data: lsd2\$LSD and lsd2\$Math

S = 162, p-value = 0.002232

alternative hypothesis: true rho is not equal to 0

sample estimates:

rho

-0.9285714

H_0 is rejected at 5% since the p -value is lower than 5%.

Kendall's τ

Yet another rank based correlation is the Kendall's tau. The principle is more complex than the two others.

- ▶ Compute the number of concordant pairs C and discordant pairs D in both samples of ranks.
- ▶ If there is no relation between X and Y , the difference between these should be small.
- ▶ If the number of concordant pairs is large, then there is a positive relationship.
- ▶ If the number of discordant pairs is large, then there is a negative relationship.

The statistics is defined as

$$r_{\tau} = \frac{C - D}{n(n - 1)/2}$$

Kendall's τ

Example:

LSD	LSD rank	Math rank	Math
1.17	1	7	78.93
2.00	2	8	150.00
2.97	3	5	58.20
3.26	4	6	67.47
4.69	5	3	37.47
5.83	6	4	45.65
6.00	7	2	32.92
6.41	8	1	29.97

The pair LSD:(1, 2) and Math:(7, 8) is concordant (both increasing). The pair LSD:(1, 3) and Math:(7, 5) is discordant (decreasing vs increasing). Etc.

In total: $C = 3$ and $D = 25$. Thus

$$r_{\tau} = (3 - 25)/(8 \times 7/2) = -0.786.$$

Kendall's τ : Why $n(n-1)/2$?

For n observations, the number of pairs is

$$n-1 + n-2 + \cdots + 2 + 1 = \frac{n(n-1)}{2}$$

Thus the statistics is bounded: $-1 \leq r_\tau \leq 1$.

Example:

Pairs including observation 1: 7

Pairs including observation 2: 6

Pairs including observation 3: 5

Pairs including observation 4: 4

Pairs including observation 5: 3

Pairs including observation 6: 2

Pairs including observation 7: 1

Total: $28 = 8 \times 7/2$

Kendall's τ test

Similar to the correlation test

$$H_0 : \rho_\tau(X, Y) = 0 \quad \text{versus} \quad H_1 : \rho_\tau(X, Y) \neq 0$$

Statistics under H_0 : $z = \frac{3(C-D)}{\sqrt{n(n-1)(2n+5)/2}} \stackrel{\sim}{\sim} N(0, 1)$.

Reject H_0 : $|z| > z_{1-\alpha/2}$

P-value: $p = 2 - 2 \times N(0, 1)(t)$.

Kendall's τ test

```
> cor.test(x=lsd2$LSD, y=lsd2$Math, method="kendall")
```

Kendall's rank correlation tau

data: lsd2\$LSD and lsd2\$Math

T = 3, p-value = 0.005506

alternative hypothesis: true tau is not equal to 0

sample estimates:

tau

-0.7857143

H_0 is rejected at 5% since the p -value is lower than 5%.

Measure of dependence: discussion

- ▶ Spearman's ρ and Kendall's τ are more robust to extreme values than Pearson's correlation.
- ▶ Pearson's correlation is more often use because of its convenient properties with normally distributed data.
- ▶ Dependence measure helps to measure dependence and to compare dependence patterns.
- ▶ They are on different scales and should not be compared: e.g. do not use the Spearman's ρ as a replacement of the Pearson's correlation unless you really know what you are doing.
- ▶ All dependence measures provide only a partial view on the dependence.

Normality and Pearson's correlation

It looks like rank based measures are better. So why is the correlation so often used?

When the variables x and y are both normally distributed then the correlation fully describes their dependence structure. In this case, the Pearson's correlation is the best possible description.

Note that:

- ▶ This is why correlation is so often used.
- ▶ This implies that there is no extreme value or that they should be put aside in some way.